

StakeSensei

Interactive sandbox where you run your own Solana validator and compare it to Ethereum

The Problem & Our Solution

Passive videos don't teach validator consensus; beginners need to click, not just watch, to grasp PoH and staking.

- Videos explain PoH, leader schedules, and stake-weighted voting once, then learners forget the mechanics.
- Ethereum's slot/epoch/attestation model adds confusion when compared only in text or narration.
- StakeSensei turns the validator lifecycle into a clickable story: stake, get selected, produce, vote, earn or lose.
- A single toggle mirrors the same scenario in Ethereum's validator/attester model for direct comparison.



How It Works

A browser sandbox simulates staking, leader selection, block production, and voting, mirrored across Solana and Ethereum.

- User flow: stake an amount, delegate, see selection odds, watch the leader rotation clock tick.
- Selected leader produces a block via simplified mock Proof of History, then other validators vote on it.
- Rewards and slashing penalties animate based on vote outcomes, reinforcing cause and effect.
- Architecture: React + Three.js/Canvas for animation, Supabase for session state, Framer Motion for transitions.
- Solana web3.js and Ethereum Beacon Chain API supply real reference data (read-only); no on-chain transactions occur.

Built with

Solana, Ethereum, React, Three.js or Canvas for animation, Solana web3.js (for real reference data), Ethereum Beacon Chain API, Supabase for session state

Why Now & What's Next

Solana's validator economics are evolving fast, and demand is growing for interactive, non-English learning tools.

- Alpenglow and Firedancer are reshaping Solana validator mechanics, raising need for up-to-date intuitive explainers.
- Japanese-language interactive content for validator consensus is scarce compared to English video explainers.
- Next: multiplayer mode where users compete as validators against each other.
- Next: guided tutorial for deploying a real testnet validator after mastering the simulation.
- Next: localize to more languages and add a quiz/certification mode.

Category

Consumer / Gaming / Education